DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2023

Course: B. Tech. **Branch: Electronics and Communication Engineering / Electronics and Telecommunication Engineering** **Semester: IV**

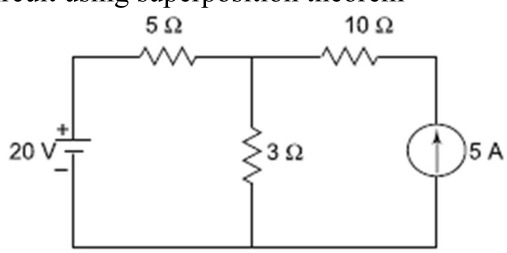
Subject Code & Name: Network Theory (BTETC401)

Max Marks: 60 **Date:16-01-24** **Duration: 3 Hr.**

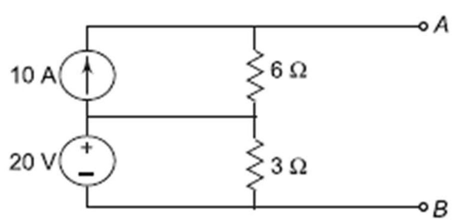
Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		**12**
A)	State Superposition theorem and find the current passing through the 3 Ω resistor in the circuit using superposition theorem	**CO1**	**6**



B)	Replace the given network shown in Fig by a single current source in parallel with a resistance	**CO1**	**6**

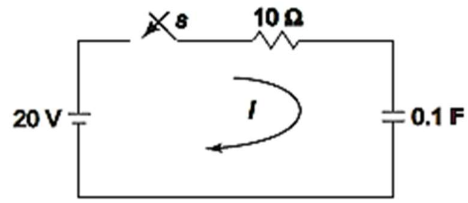


C)	Define the terms: TREE AND CO-TREE, Obtain the incidence matrix A from the following reduced incidence matrix A1 and draw its graph.	**CO1**	**6**

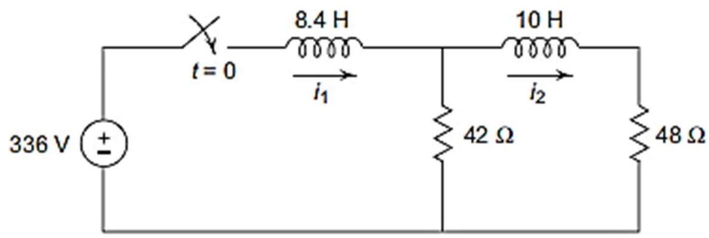
$$[A_1] = \begin{bmatrix} -1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & -1 & 1 \end{bmatrix}$$

Q.2	**Solve Any Two of the following.**		**12**

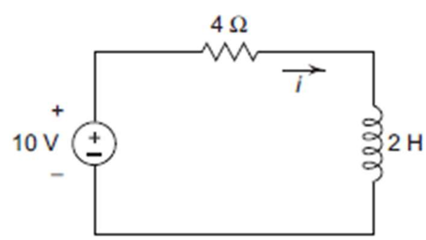
A)	A series RC circuit consists of resistor of 10 Ω and capacitor of 0.1 F as shown in Fig. A constant voltage of 20 V is applied to the circuit at t = 0. Obtain the current equation. Determine the voltages across the resistor and the capacitor.	**CO1, CO3**	**6**



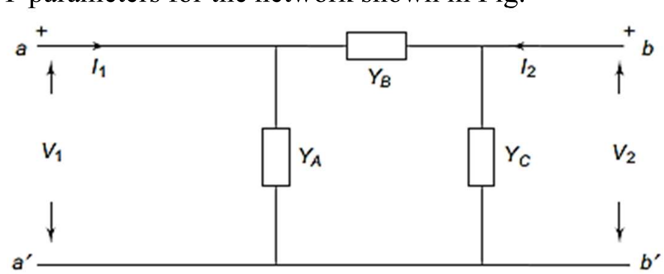
B)	Derive an expression for Dc response of an R-L circuit	**CO1, CO3**	**6**
C)	Obtain the expression for i_1 and i_2 in the circuit shown in Fig. when dc voltage source is applied suddenly. Assume that the initial energy stored in the circuit is zero.	**CO1, CO3**	**6**



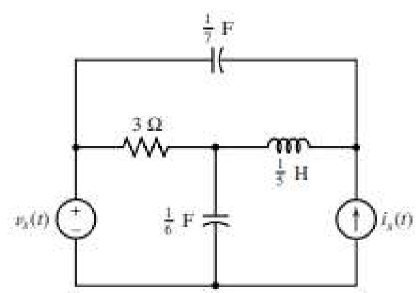
Q. 3	**Solve Any Two of the following.**		**12**
A)	Write Laplace transform of some standard Network Functions.	**CO1, CO3**	**6**
B)	Explain Circuit elements in the s-domain	**CO1, CO3**	**6**
C)	Determine the current i for t ≥ 0 if initial current i(0) = 1 for the circuit shown in Fig.	**CO1**	**6**



Q.4	**Solve Any Two of the following.**		**12**
A)	Find the Y parameters for the network shown in Fig.	**CO1, CO4**	**6**



B)	In the two-port network, compute h-parameters from the following data: (a) With the output port short-circuited: $V1 = 25$ V, $I1 = 1$ A, $I2 = 2$ A (b) With the input port open-circuited: $V1 = 10$ V, $V2 = 50$ V, $I2 = 2$ A	**CO1, CO4**	**6**
C)	Derive Z-parameter in terms of Y-parameter, h-parameter and ABCD parameter	**CO1, CO4**	**6**
Q. 5	**Solve Any Two of the following.**		**12**
A)	Find the two Cauer realisations of driving point function given by $$Z(S) = \frac{10S^4 + 12S^2 + 1}{2S^3 + 2S}$$	**CO1, CO3**	**6**
B)	Obtain the normal-form equations for the circuit of Fig., a four-node circuit containing two capacitors, one inductor, and two independent sources.	**CO1, CO3**	**6**



C)	Explain about classification of filters. Draw the characteristics of Low-pass filters.	**CO1, CO3**	**6**

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